

Problem

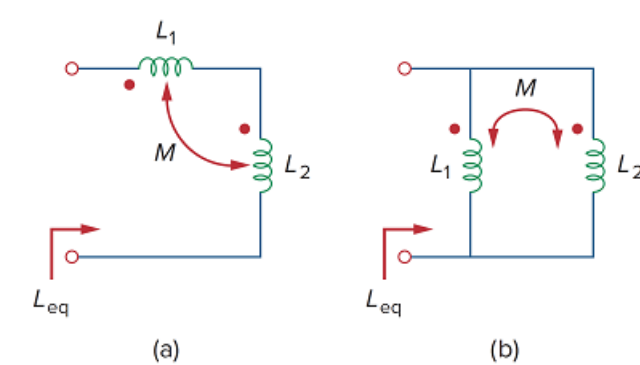
(a) For the coupled coils in Fig. 13.74(a), show that

$$L_{eq} = L_1 + L_2 + 2M$$

(b) For the coupled coils in Fig. 13.74(b), show that

$$L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M}$$

Figure 13.74



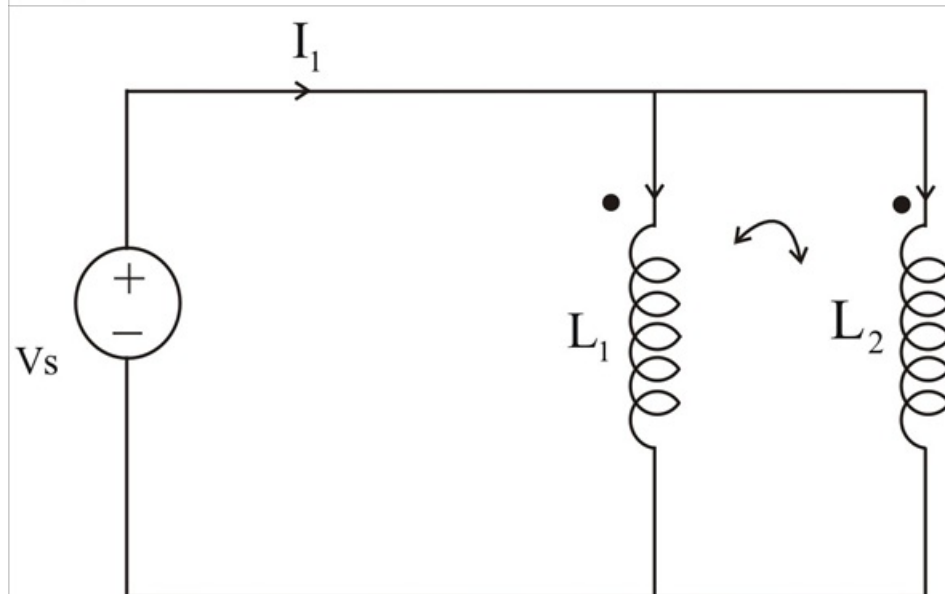
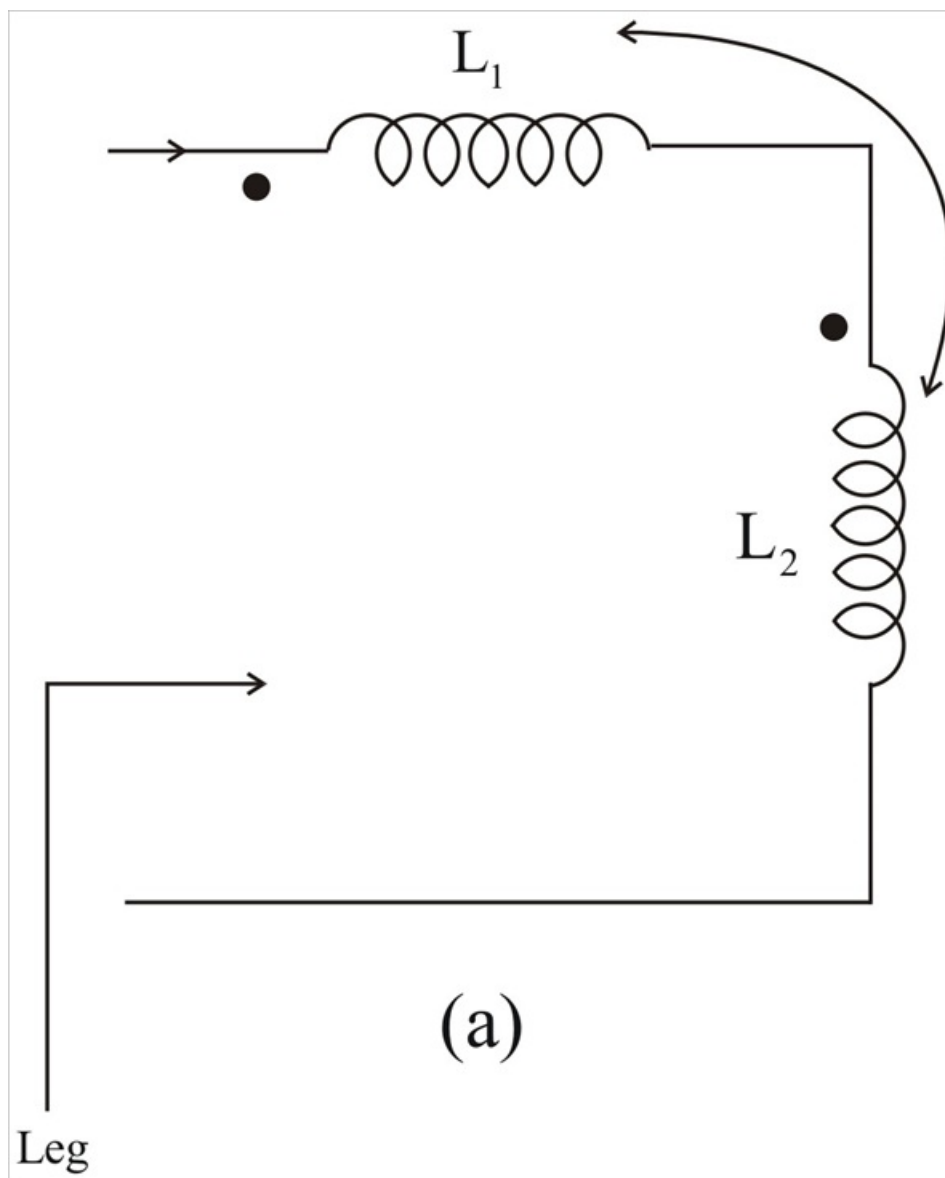
Step-by-step solution

Step 1 of 3

(A) For the series connections shown in figure (a). The current i enters each coil from its dotted terminal. Therefore, the mutually induced voltages have the same sign as the self induced voltages. Thus,

$$L_{eq} = L_1 + L_2 + 2M$$

Step 2 of 3



(B) For the parallel coil, consider figure (b).

$$I_s = I_1 + I_2 \quad \text{And} \quad Z_{eq} = \frac{V_s}{I_s}$$

Applying KVL to each branch gives,

$$V_s = j\omega L_1 I_1 + j\omega M I_2 \rightarrow (1)$$

$$V_s = j\omega M I_1 + j\omega L_2 I_2 \rightarrow (2)$$

Or,

$$\begin{bmatrix} V_s \\ V_s \end{bmatrix} = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$\Delta = -\omega^2 L_1 L_2 + \omega^2 M^2 ,$$

$$\Delta_1 = j\omega V_s (L_2 - M) ,$$

$$\Delta_2 = j\omega V_s (L_1 - M)$$

$$I_1 = \frac{\Delta_1}{\Delta} , \quad \text{And} \quad I_2 = \frac{\Delta_2}{\Delta}$$

$$I_s = I_1 + I_2 = \frac{(\Delta_1 + \Delta_2)}{\Delta}$$

$$= \frac{j\omega(L_1 + L_2 - 2M)V_s}{[-\omega^2(L_1 L_2 - M^2)]}$$

$$= \frac{(L_1 + L_2 - 2M)V_s}{[j\omega(L_1 L_2 - M^2)]}$$

$$Z_{eq} = \frac{V_s}{I_s}$$

$$= \frac{j\omega(L_1 L_2 - M^2)}{(L_1 + L_2 - 2M)}$$

$$= j\omega L_{eq}$$

i.e.,

$$L_{eq} = \frac{(L_1 L_2 - M^2)}{(L_1 + L_2 - 2M)} .$$